SAMAY SARTHAK SWAIN

SamaySarthakSwain | Samay Sarthak Swain | samaysarthakswain@gmail.com | +91 8327780375

Summary

Dynamic Full Stack Developer specializing in AI-integrated web applications and scalable system architecture, with hands-on experience building innovative solutions such as AI-driven healthcare systems and smart IoT projects. Currently strengthening skills in AWS, Data Structures, and cloud technologies to develop efficient, high-performance applications. Proven problem solving and leadership abilities through active participation in hackathons and technical projects, and driven by a strong interest in solving real-world problems through user centric and impactful technology.

Work Experience

UI/UX & Graphic Designer – MMG Graphics Oct 2023 - Jan 2024

Handled brand identity and digital interface designs.

Technical Research Intern – NIST University May 2024 - Jul 2024

– Worked on Object-Oriented Programming (OOPS) concepts in C++

Projects

Smart Crowd & Traffic Management System (AI + IoT Platform) Jan 2026 - Present

- Developed AI-based system for real-time crowd and traffic monitoring
- Integrated multi-camera support across CCTV, IP cameras, and edge devices
- Implemented computer vision for vehicle detection and crowd density analysis
- Enabled intelligent traffic control with congestion prediction

Lets Explore – AI Tourism Platform May 2025 - Present

- Built AR/VR-based monument exploration using React.js and Three.js with 3D models
- Integrated Web Speech API for voice-based navigation
- Developed AI travel assistant using LLMs (Gemini/OpenAI)
- Implemented Emergency SOS using Google Maps API

NeuroGuard – AI Healthcare Web Platform Feb 2025 - Present

- Developed full-stack healthcare monitor using Node.js, Express, FastAPI
- Built AI-based real-time health monitoring system
- Designed user-friendly UI using Tailwind CSS

Ear-to-Heart – AI-Based Heart Disease Prediction System 2026

- Developed a wearable, non-invasive system for early cardiovascular risk prediction using ear-based physiological signals
- Integrated ECG and PPG sensing with supporting temperature and motion measurements
- Extracted heart rate, HRV, pulse, and ECG/PPG waveform features for machine-learning prediction
- Evaluated classification approaches including Logistic Regression, SVM, Random Forest, and XGBoost

V2V Communication & Vehicle Black Box System Oct 2025

- Designed V2V communication system using C++ and MQTT
- Built hardware system using Arduino/ESP32
- Enabled real-time accident data logging for emergency response

Ear-to-Heart Disease Prediction Aug 2026

- Proposed an AI-based wearable system for early prediction of heart disease using ear-based physiological signals
- Processed ECG and PPG signals to extract features like heart rate, HRV, and pulse characteristics
- Applied ML models (Logistic Regression, SVM, Random Forest, XGBoost) for risk classification
- Designed a connected application interface for portable, preliminary cardiovascular risk assessment

Education

2024 - 2028	B.Tech in Computer Science (AI/ML) at **NIST University**	Overall CGPA : 8.8
2022 - 2024	Class 12th – KC Public School	(80%)
2020 - 2022	Class 10th – KC Public School	(93%)

Skills

Frontend & Design	React.js, Three.js, JavaScript, HTML, CSS, Tailwind CSS, UI/UX
Backend & Languages	TypeScript, Node.js, FastAPI, Python, Java, C/C++
Database	MySQL, MongoDB
Tools & Infra	Git, GitHub, Docker, Netlify, Vercel, REST APIs

Achievements

- GDG Atlas OnCampus – Winner
- Odisha AI Symposium – 1st Runner-up
- Startup Odisha Campus Hackfest – Finalist
- Techspire (NIT Bhubaneswar) – Finalist
- Git & GitHub Workshop – 2nd Runner-up